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# ChronoQA

*A Large-Scale Timestamp-Anchored Multiple-Choice QA Training Dataset for Training Temporally Grounded Video-Language Models*

| 40,000+ | 5–60 min | 5 Categories | 5-way MCQ | ~20% | Dual | ~98% |
|---|---|---|---|---|---|---|
| SOURCE VIDEOS | VIDEO DURATION RANGE | TEMPORAL REASONING TYPES | PER EXAMPLE FORMAT | TEXT-ONLY SOLVABILITY TARGET | SUPERVISION SIGNAL | PIPELINE SUCCESS RATE |

## MOTIVATION

Large multimodal language models (MLLMs) have achieved impressive performance on standard video benchmarks, but their high scores conceal a fundamental weakness: **temporal grounding**, the ability to localize *when* events occur, not merely *what* they are.

This gap exists because current training data provides answer supervision but completely lacks **temporal evidence supervision**. Models learn which answer is correct, but never learn where in the video that answer is supported.

> **Gemini 3 Pro** scores 88.4% on Video-MME yet achieves only **48.2%** on fine-grained temporal tasks, while humans score **98.2%**. (TimeBlind, 2026)

Without explicit grounding supervision, models exploit shortcuts instead of learning genuine temporal reasoning:

1. **Static Shortcuts**: Answering from single-frame cues that happen to correlate with correct answers, without engaging the video's temporal structure.
2. **Linguistic Priors**: Pattern-matching question text to answer options without watching the video at all.
3. **Positional Bias**: Exploiting the tendency of existing datasets to cluster evidence near video boundaries, enabling imprecise localization to score well.

Practical applications like surgical procedure review, broadcast sports analysis, security monitoring, and long-form lecture comprehension, all require models to answer *and* to identify precisely when supporting evidence occurs. ChronoQA addresses this gap at the training data level.

## THE GAP IN EXISTING DATASETS

No existing training resource combines long-form video, MCQ format, and timestamp-level grounding targets simultaneously:

| Dataset | Long-form | Timestamp Labels | MCQ | Training Signal |
|---|---|---|---|---|
| **Video-MME** | ✓ | ✗ | ✓ | Answer Only |
| **Charades-STA** | ✗ | ✓ | ✗ | Regression / Biased |
| **ActivityNet-QA** | ~3 min | ✗ | ✗ | Answer Only |
| **LongVideoBench** | ✓ | Input | ✓ | Inverted Task |
| **MVBench** | ✗ | Coarse Bins | ✓ | Answer Only |
| **EgoSchema** | 3 min | ✗ | ✓ | Answer Only |
| **ChronoQA** | ✓ | ✓ | ✓ | **Answer + Interval** |

*LongVideoBench provides timestamps as input context, an inversion of the task. ChronoQA requires models to independently locate the supporting evidence interval as a supervision target.*

## DATASET DESIGN

Each ChronoQA training instance provides three simultaneous levels of supervision, enabling joint training on answer correctness and temporal localization quality.

| **Answer** | $[t_s, t_e]$ | **4 Distr-actors** |
|---|---|---|
| Correct label A – E via cross-entropy | Verified timestamp interval per answer | Semantically plausible, yet wrong options |

> **No Temporal Leakage:** Questions never reference "when," "at the beginning," or any timestamp/temporal cues. Blind-filtered so text-only models cannot exceed ~20% (random baseline) without visual context, ensuring every example genuinely requires visual temporal engagement to answer correctly.

## EXAMPLE TRAINING INSTANCE

QUESTION (ACTION UNDERSTANDING)

**"While the chef is working at the wooden cutting board, what is he doing with the herbs immediately before adding them to the pan?"**

ANSWER OPTIONS

- A Roughly tearing the herbs by hand into uneven pieces
- B Blanching the herbs briefly in boiling water to wilt them
- C Finely chopping the herbs using a rocking motion with a chef's knife
  7:08 – 7:21
- D Bruising the herb stems against the board with the flat of the blade
- E Patting the herbs dry with a cloth before loosely stacking them

**Training targets:**
Label = C | Grounding interval = [7:08, 7:21]

*Distractors describe real events from **elsewhere in the same video**. Models must locate the correct segment, not just recognize what seems like a plausible description.*

## FIVE TRAINING CATEGORIES

*In Order of by Temporal Scope Required:*
*Segment-level to Full-video*

1. **Action Understanding**
   Fine-grained characterization of a video segment's content. Includes tool usage, body posture, object manipulation, and procedural steps within a single continuous action.
   *Scope: Single Segment*
   *Tests: Visual Perception*
2. **Needle-in-Haystack**
   Retrieval of specific visual or textual details. Includes brand names, colors, on-screen text, counts, spatial attributes, from anywhere in the full video. Correct-answer timestamps uniformly distributed to prevent positional bias.
   *Scope: Any Point in the Video*
   *Tests: Sustained Attention*
3. **Temporal Ordering**
   Reasoning about before/after relationships, event sequences, and transitions. Events may be separated by minutes, requiring a coherent temporal map of the full video.
   *Scope: Multi-segment*
   *Tests: Sequence Reasoning*
4. **Causal Understanding**
   Integrates evidence from two or more non-adjacent segments to infer causation, motivation, and cross-segment patterns. No single segment contains sufficient information to answer correctly.
   *Scope: Multiple Disjoint Segments*
   *TTests: Cross-segment Synthesis*
5. **General Understanding**
   Holistic comprehension of the video's overarching narrative, primary subject, core message, or tone. Correct answers are only derivable after processing the video in its entirety.
   *Scope: Full Video*
   *Tests: Global Narrative Reasoning*

## AUTOMATED GENERATION PIPELINE

ChronoQA is produced by a fully automated pipeline, requiring no manual annotation, enabling scaling to hundreds of thousands of training examples.

**1. YouTube Crawl**
*Filter by duration (5–60 min), content type (instructional, narrative, procedural, sports, vlogs), and Creative Commons licensing.*

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**2. Visual Diversity Filter**
*Compute per-video delta score from CLIP embeddings & cosine similarity across sampled frames. Discard low-motion, static-content videos.*

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**3. Segmentation**
*Split each video into fixed 10s clips. Experiments with 5s, 10s, 20s, and 30s segments show 10s strikes the optimal balance of temporal resolution vs. computational cost.*

↓

**4. Caption Generation**
*Qwen3-VL-30B via vLLM at $1 FPS with token budget. Each segment yields a timestamped free-form caption. Token budget determines representation space for visual detail per frame-pair.*

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**5. QA Generation**
*Qwen3-Next-80B-A3B-Thinking with Pydantic-enforced structured JSON. Five category-specific prompt templates, with 3 questions per category per video. ~98% generation success rate.*

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**6. Validation & Cleaning**
*Pattern matching removes time-referencing questions. Adjacent timestamps <2s apart are merged. Timestamp jitter of random(1,3)s applied to start and end to increase annotation variance.*

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**7. Distractor Generation**
*Separate LLM pass generates 4 plausible distractors per question, grounded in ±30s local caption context. Distractors match the correct answer in length, casing, and semantic category.*

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**8. Blind Filtering**
*Multiple text-only LLMs (including Qwen3-Next-80B) are presented with question + options only. Questions answerable by multiple models are discarded. This ensures visual necessity of every retained example.*

## CAPTION TOKEN BUDGET

Visual token count per video segment is controlled via the $T_{\text{video}}$ parameter, which caps total pixel input across all sampled frames:

*Pixel cap and per frame-pair budget*

$$P_{\text{video}} = T_{\text{video}} \times 32^2 = T_{\text{video}} \times 1{,}024$$

$$T_{\text{fp}} = T_{\text{video}}/t \quad \text{(tokens per frame-pair)}$$

At $1FPS$ with temporal compression factor of 2, a $t$-second segment produces $\frac{t}{2}$ frame-pairs. Higher $T_{\text{video}}$ yields finer detail but increases KV-cache pressure and reduces inference concurrency via PagedAttention. Token budgets of 32, 64, and 128 tokens per frame-pair were evaluated. 64 was selected as the production setting.

## TRAINING OBJECTIVE

Each ChronoQA example supports a dual training objective, jointly optimizing answer correctness and temporal localization quality:

*Answer Supervision: Cross-entropy*

$$\mathcal{L}_{\text{ans}} = \text{CrossEntropy}(\hat{a},\, a^*)$$

$$\text{where } a^* \in \{A, B, C, D, E\}$$

*Temporal Grounding: IoU-based loss*

$$\text{tIoU} = \frac{|P \cap G|}{|P \cup G|} \times 100\%$$

$$\mathcal{L}_{\text{grd}} = 1 - \text{tIoU}(\text{predicted},\, [t_s,\, t_e])$$

*Joint objective*

$$\mathcal{L} = \mathcal{L}_{\text{ans}} + \lambda \cdot \mathcal{L}_{\text{grd}}$$

> A model trained on answer labels alone can reach high accuracy by exploiting shortcuts. The grounding loss forces the model to internalize *where* the answer is supported, producing representations that generalize to genuinely novel temporal queries.

Grounded Accuracy (G-Acc) measures joint success: A correct answer *and* $\text{tIoU} \geq 0.5$ on the predicted interval, providing a unified metric for downstream evaluation of trained models.

## QUALITY CONTROL

- ✓ **Blind Filtering**: Performance of multiple text-only LLMs cannot exceed random chance. Visual engagement is structurally required, not assumed.
- ✓ **Timestamp Jitter**: ±1–3s random expansion of ground-truth intervals prevents models from memorizing exact 10s clip boundaries.
- ✓ **Uniform Timestamp Distribution**: Needle-in-Haystack correct-answer timestamps are placed evenly across the full video timeline, eliminating start/end positional bias found in previous works.
- ✓ **Balanced Answer Positions**: Correct answers are uniformly distributed across $A - E$, preventing learned position shortcuts.
- ✓ **Visual Diversity Filtering**:A CLIP-based delta score removes static, low-motion videos to ensure that every example demands temporal attention.
- ✓ **Semantically Grounded Distractors**: Wrong options are real events that have occurred in the same video, forcing contrastive temporal discrimination rather than pattern matching.

## DOWNSTREAM USE CASES

**Supervised Fine-Tuning**
Train VLMs on answer + timestamp pairs to instill temporal grounding as an explicit learned capability on top of pretrained foundations.

**RLHF / Preference Data**
Use grounded vs. ungrounded correct answers as preference pairs for reward modeling. Grounded answers preferred over correct, but unlocalized, ones.

**Pretraining Augmentation**
Mix into large-scale video pretraining corpora to improve temporal awareness from the ground up, prior to task-specific fine-tuning.

**Probing & Diagnostics**
Held-out split supports controlled evaluation of temporal grounding across all five reasoning categories and video duration ranges.

**Curriculum Learning**
Five categories naturally ordered by difficulty, training on Action Understanding first, and progressing towards Causal and General as temporal reasoning matures.

**Domain Adaptation**
Instructional and procedural video subsets directly support domain-specific fine-tuning for medical, legal, and educational video applications.

## VLM ARCHITECTURE & TEMPORAL BOTTLENECK

**Video Frames** (Sampled at $N$ fps) → **Vision Encoder** (SigLIP-2 / InternViT) → **Patch Tokens** (Projected to LLM space) → **LLM Backbone** (Generates Response)

⚠ Inter-frame temporal signals are discarded at projection. Thus, motion fidelity is entirely sampling-density dependent.

**Proprietary**
Gemini 2.5 Pro: 2M token context length allows for ~6h @ 1 fps. Performance still degrades with duration, meaning context length alone is insufficient.

**Open-source**
Qwen3-VL, Molmo2: Introduce spatiotemporal grounding but operate at lower effective temporal resolution.

> ChronoQA's dual supervision directly targets this existing bottleneck , providing the explicit temporal evidence signal that existing pretraining corpora lack, enabling models to learn to ground rather than takes shortcuts around it.

## DISTRACTOR DESIGN PHILOSOPHY

Each question receives exactly 4 wrong answers generated by a dedicated LLM pass using a specialist prompt. Distractors are designed to be plausible to an inattentive viewer while remaining clearly incorrect:

- **Same-category Constraint:** Distractors belong to the same semantic category as the correct answer. If the answer is a color, all distractors are colors. If an action, all distractors describe actions. This prevents trivial elimination by type.
- **Scene-grounded Alternatives:** Distractors are drawn from the same scene. Similar objects, nearby people, or adjacent details that are present but not the correct answer. They must feel like they *could* have been the answer given the same scene.
- **Anti-shortcut Formatting:** All distractors match the casing style and approximate length of the correct answer. Length asymmetry is a well-known test-taking cue that allows correct guessing without visual understanding.

## CONTRIBUTIONS

1. **ChronoQA Training Corpus**: The first large-scale dataset pairing long-form video QA with verified timestamp supervision across five temporal reasoning categories, sourced from 40,000+ YouTube videos of 5–60 minutes duration.
2. **Scalable Automated Pipeline**: Fully automated generation from crawl to quality-filtered training examples via Qwen3-VL captioning, structured QA generation, and multi-model blind filtering. No manual annotation required.
3. **Dual Training Objective**: Answer & grounding supervision per instance directly targets the gap between QA accuracy and temporal localization ability, enabling models trained on ChronoQA to be evaluated with the joint G-Acc metric.
4. **Principled Quality Controls**: Blind filtering, timestamp jitter, uniform distribution, and visual diversity filtering together ensure every training example is visually necessary, positionally unbiased, and resistant to linguistic shortcuts.

Video-Language Models · Temporal Grounding · Training Data · Long-Form Video · Multimodal Learning

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